



Costa Rica Institute of Technology

Escuela de Ingeniería en Computación

Artificial Intelligence • IC-6200

Semester Project

Development log across Stages 1, 2 and 3

Applied AI Agent Project — Work Journal

Decisions, blockers, agreements and milestones

Work journal • Sprint log • Technical decisions • Traceability

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1 Purpose and Conventions

1.1 Purpose

This journal records the development of the project across its three stages: what was decided, why, what blocked progress, and what evidence backs each claim. It is not the paper and it is not the sprint plan. The paper reports results; the sprint plan states intent; this document records what actually happened.

1.2 Team and Repository

Table 1.1: Team members and primary responsibilities.

Member	Review axis	Primary responsibility
Adriel S. Chaves Salazar	Axis B — techniques	[TO BE COMPLETED: responsibility]
Daniel Duarte Cordero	Axis A — problem	[TO BE COMPLETED: responsibility]
Sebastián Hernández Bonilla	Axis C — dataset	[TO BE COMPLETED: responsibility]

Repository: `Applied-AI-Agent-Project`. Branching and commit conventions live in `CodeControl.md`; the sprint backlog lives in `SprintPlanning.md`.

1.3 How Entries Are Written

Every entry opens with a dated header naming the owner and the branch the work happened on. Inside an entry, three callouts carry the weight:

Decision. The choice that was made.

Rationale. Why it was made, with evidence.

Discarded. The alternatives considered and the reason each was dropped.

Blocker. What stopped progress.

Impact. What it delayed and by how much.

Resolution. How it was resolved, or the condition that will resolve it.

Milestone. What was achieved.

Evidence. Where the proof lives: commit, log, figure, screenshot.

Writing rule. Write entries the day the work happens. An entry written a week later loses the reasoning, which is the only part of this document that cannot be reconstructed from the repository history.

1.4 Traceability

Every decision recorded here that affects a technical choice must appear, in condensed form, in the corresponding section of the paper. Every milestone must point at a committed artifact. AI tool usage is annotated inline and consolidated in `AI_USAGE.md`.

2 Stage 1 — State of the Art and Design

2.1 Sprint Summary

Table 2.1: Stage 1 branches and their outcome.

Branch	Window	Owner	Outcome
00TrackConfirmation	21/08	A. Chaves	[TO BE COMPLETED: outcome]
01LiteratureReview	22/08–23/08	S. Hernández	[TO BE COMPLETED: outcome]
02TechnicalDesign	24/08–25/08	A. Chaves	[TO BE COMPLETED: outcome]
03PaperAndDelivery	26/08–27/08	D. Duarte	[TO BE COMPLETED: outcome]

2.2 Log

2026-08-21 • Track approval request and repository bootstrap

Owner: A. Chaves, S. Hernández, D. Duarte • Branch: 00TrackConfirmation

[TO BE COMPLETED: What was done. Two to four sentences, factual.]

Decision. [TO BE COMPLETED: the decision]

Rationale. [TO BE COMPLETED: the rationale]

Discarded. [TO BE COMPLETED: alternatives dropped]

Milestone. [TO BE COMPLETED: what was achieved]

Evidence. [TO BE COMPLETED: commit hash, file, or screenshot]

AI usage: tool: [TO BE COMPLETED: tool] • purpose: [TO BE COMPLETED: purpose] •
verified: [TO BE COMPLETED: what was verified manually]

2026-08-22 • Three-axis literature review — first pass

Owner: S. Hernández, D. Duarte, A. Chaves • Branch: 01LiteratureReview

[TO BE COMPLETED: What was done.]

Blocker. [TO BE COMPLETED: what stopped progress]

Impact. [TO BE COMPLETED: impact]

Resolution. [TO BE COMPLETED: resolution]

2026-08-23 • Comparative table and problem formalization

Owner: [TO BE COMPLETED: owner] • Branch: 01LiteratureReview

[TO BE COMPLETED: What was done.]

2026-08-24 • Track technical analysis and literature baseline

Owner: [TO BE COMPLETED: owner] • Branch: 02TechnicalDesign

[TO BE COMPLETED: What was done.]

2026-08-25 • Experimental protocol, baselines, risks and ethics

Owner: [TO BE COMPLETED: owner] • Branch: 02TechnicalDesign

[TO BE COMPLETED: What was done.]

2026-08-26 • Paper assembly, AI declaration, content freeze

Owner: [TO BE COMPLETED: owner] • Branch: 03PaperAndDelivery

[TO BE COMPLETED: What was done.]

2026-08-27 • Final review, merge to master, submission

Owner: [TO BE COMPLETED: owner] • Branch: 03PaperAndDelivery

[TO BE COMPLETED: What was done.]

2.3 Consolidated Decisions

Table 2.2: Technical decisions taken during Stage 1.

Date	Decision	Rationale	Paper section
[TO BE COMPLETED: date]	[TO BE COMPLETED: decision]	[TO BE COMPLETED: rationale]	[TO BE COMPLETED: section]

2.4 Retrospective

Table 2.3: Stage 1 retrospective.

Question	Answer
What went well?	[TO BE COMPLETED: answer]
What blocked us?	[TO BE COMPLETED: answer]
Which estimates were wrong?	[TO BE COMPLETED: answer]
What changes for Stage 2?	[TO BE COMPLETED: answer]

3 Stage 2 — Modeling and Evaluation

Status. Planned at the Stage 1 retrospective. Scope: complete methodological cycle — data pipeline, internal baselines, main model, ablations, experiment tracking, and comparable result tables against the three references.

3.1 Sprint Summary

[TO BE COMPLETED: Branch table, filled once the sprint is planned.]

3.2 Log

[TO BE COMPLETED: Dated entries, added as the work happens.]

3.3 Experiment Register

Table 3.1: Experiment runs and their outcome.

Run ID	Configuration	Seed	Metric	Observation
[TO BE COMPLETED: id]	[TO BE COMPLETED: config]	[TO BE COMPLETED: seed]	[TO BE COMPLETED: value]	[TO BE COMPLETED: note]

3.4 Consolidated Decisions

[TO BE COMPLETED: Table, same columns as Stage 1.]

3.5 Retrospective

[TO BE COMPLETED: Filled at the end of the stage.]

4 Stage 3 — Agent and Final Paper

Status. Planned at the Stage 2 retrospective. Scope: functional intelligent agent consuming the model in a verifiable way, deployment ethics, and integration of Stage 2 results into the final paper.

4.1 Sprint Summary

[TO BE COMPLETED: Branch table.]

4.2 Log

[TO BE COMPLETED: Dated entries.]

4.3 Agent Integration Notes

[TO BE COMPLETED: Interface exposed by the model, tools available to the agent, and how consumption is verified.]

4.4 Consolidated Decisions

[TO BE COMPLETED: Table, same columns as Stage 1.]

4.5 Final Retrospective

[TO BE COMPLETED: Filled at the end of the project.]

A Meeting Minutes

A.1 Index

Table A.1: Meetings held during the project.

Date	Duration	Attendees	Main topic
[TO BE COMPLETED: date]	[TO BE COMPLETED: duration]	[TO BE COMPLETED: attendees]	[TO BE COMPLETED: topic]

A.2 Minutes

2026-08-[TO BE COMPLETED: DD] • [TO BE COMPLETED: meeting title]

Owner: [TO BE COMPLETED: attendees] • Branch: --

Agenda.

- [TO BE COMPLETED: item]

Discussion. [TO BE COMPLETED: What was actually argued, including positions that did not win.]

Agreements.

- [TO BE COMPLETED: agreement, with owner and deadline]

Open questions.

- [TO BE COMPLETED: question]